

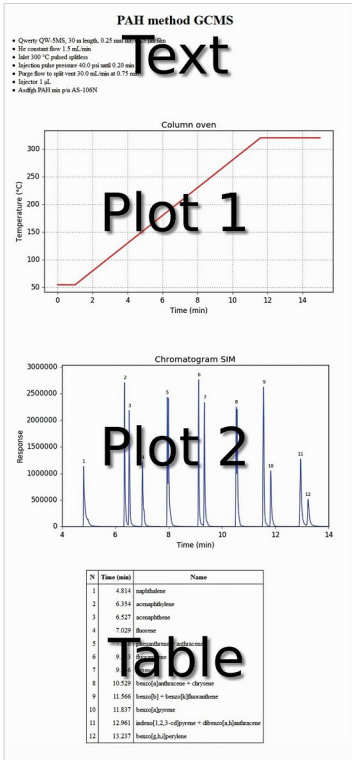


Figure 1: Python and org-mode

the system. You can do this by going to (Windows 7) *Start* → *Control panel* → *System* → *Advanced system settings* → *Environment variables* → *User variables for <your-username>* → *Create*, if not present, or modify the variable path → Add C:\WinPython-32bit-3.4.4.2\python-3.4.4;

Another method is to use Emacs as an IDE. A Python file can simply be evaluated by pressing the F8 function key (see the above mentioned *kbd "<f8>"* option). Figure 2 shows an Emacs session with three buffers opened. On the left side is the Python code, on the right side on the top

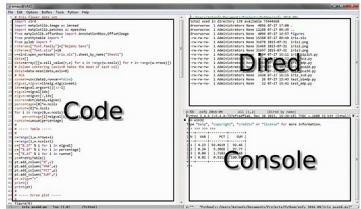


Figure 2: Python and Emacs

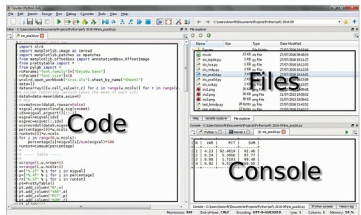


Figure 3: Spyder IDE

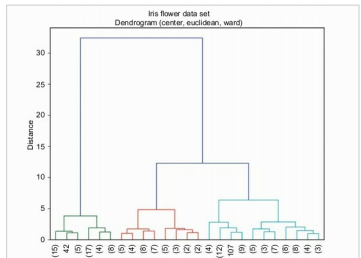


Figure 4: Agglomerative hierarchical clustering

a dired buffer as file manager and on the right side bottom is the Python console with a tabular output. This is not very different from the Spyder IDE (which is included in the WinPython distribution) shown in Figure 3, with the same three buffers opened.

Hierarchical cluster analysis

This example is about agglomerative hierarchical clustering. The data table is the famous iris flower data set and is taken from http://en.wikipedia.org/wiki/Iris_flower_data_set. It has 150 rows and five columns: sepal length, sepal width, petal length, petal width, species' name (iris setosa from row 1 to